Practical – 7

Write a program that the addresses of code, global, static, heap, and stack variables. Analyse the Linux process address space layout and explain each memory segment.

```
  GNU nano 8.7.1                          memory.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global = 100;
static int s = 50;

int main()
{
    int stack = 10;
    int *heap = (int *)malloc(sizeof(int));

    printf("Process ID     : %d\n", getpid());
    printf("Code Segment   : %p\n", (void *)main);
    printf("Global Segment : %p\n", (void *)&global);
    printf("Static Segment : %p\n", (void *)&s);
    printf("Heap Segment   : %p\n", (void *)heap);
    printf("Stack Segment  : %p\n", (void *)&stack);

    getchar();

    free(heap);

    return 0;
}
```

```
ketana_sindhu@KetaZone:~$ nano memory.c
ketana_sindhu@KetaZone:~$ gcc memory.c -o memory
ketana_sindhu@KetaZone:~$ ./memory
Process ID     : 711
Code Segment   : 0x5a889645b1e9
Global Segment : 0x5a889645e010
Static Segment : 0x5a889645e014
Heap Segment   : 0x5a88b8884010
Stack Segment  : 0x7ffd52b947ec
```

```
ketana_sindhu@KetaZone:~$ ps
    PID TTY          TIME CMD
    349 pts/0    00:00:00 bash
    784 pts/0    00:00:00 ps
ketana_sindhu@KetaZone:~$
```

```
ketana_sindhu@KetaZone:~$ cat /proc/349/maps
635b3ec75000-635b3eca6000 r--p 00000000 08:30 1503                       /usr/bin/bash
635b3eca6000-635b3eda9000 r-xp 00031000 08:30 1503                       /usr/bin/bash
635b3eda9000-635b3ede0000 r--p 00134000 08:30 1503                       /usr/bin/bash
635b3ede0000-635b3ede4000 r--p 0016b000 08:30 1503                       /usr/bin/bash
635b3ede4000-635b3eded000 rw-p 0016f000 08:30 1503                       /usr/bin/bash
635b3eded000-635b3edf9000 rw-p 00000000 00:00 0
635b76ac1000-635b76c2c000 rw-p 00000000 00:00 0                          [heap]
78ae06da4000-78ae06dff000 r--p 00000000 08:30 2734                       /usr/lib/locale/C.utf8/LC_CTYPE
78ae06dff000-78ae06e00000 r--p 00000000 08:30 2741                       /usr/lib/locale/C.utf8/LC_NUMERIC
78ae06e00000-78ae06e28000 r--p 00000000 08:30 14176                      /usr/lib/x86_64-linux-gnu/libc.so.6
78ae06e28000-78ae06fc0000 r-xp 00028000 08:30 14176                      /usr/lib/x86_64-linux-gnu/libc.so.6
78ae06fc0000-78ae0700e000 r--p 001c0000 08:30 14176                      /usr/lib/x86_64-linux-gnu/libc.so.6
78ae0700e000-78ae07012000 r--p 0020d000 08:30 14176                      /usr/lib/x86_64-linux-gnu/libc.so.6
78ae07012000-78ae07014000 rw-p 00211000 08:30 14176                      /usr/lib/x86_64-linux-gnu/libc.so.6
78ae07014000-78ae07021000 rw-p 00000000 00:00 0
78ae07021000-78ae07022000 r--p 00000000 08:30 2744                       /usr/lib/locale/C.utf8/LC_TIME
78ae07022000-78ae07023000 r--p 00000000 08:30 2733                       /usr/lib/locale/C.utf8/LC_COLLATE
78ae07023000-78ae07024000 r--p 00000000 08:30 2739                       /usr/lib/locale/C.utf8/LC_MONETARY
78ae07024000-78ae07025000 r--p 00000000 08:30 2738                       /usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC
_MESSAGES
78ae07025000-78ae07026000 r--p 00000000 08:30 2742                       /usr/lib/locale/C.utf8/LC_PAPER
78ae07026000-78ae0702d000 r--s 00000000 08:30 13860                      /usr/lib/x86_64-linux-gnu/gconv/gconv-mod
ules.cache
78ae0702d000-78ae07030000 rw-p 00000000 00:00 0
78ae07030000-78ae0703e000 r--p 00000000 08:30 14514                      /usr/lib/x86_64-linux-gnu/libtinfo.so.6.6
78ae0703e000-78ae07052000 r-xp 0000e000 08:30 14514                      /usr/lib/x86_64-linux-gnu/libtinfo.so.6.6
78ae07052000-78ae07060000 r--p 00022000 08:30 14514                      /usr/lib/x86_64-linux-gnu/libtinfo.so.6.6
78ae07060000-78ae07064000 r--p 0002f000 08:30 14514                      /usr/lib/x86_64-linux-gnu/libtinfo.so.6.6
78ae07064000-78ae07065000 rw-p 00033000 08:30 14514                      /usr/lib/x86_64-linux-gnu/libtinfo.so.6.6
78ae07065000-78ae07066000 r--p 00000000 08:30 2740                       /usr/lib/locale/C.utf8/LC_NAME
78ae07066000-78ae07067000 r--p 00000000 08:30 2732                       /usr/lib/locale/C.utf8/LC_ADDRESS
78ae07067000-78ae07068000 r--p 00000000 08:30 2743                       /usr/lib/locale/C.utf8/LC_TELEPHONE
78ae07068000-78ae07069000 r--p 00000000 08:30 2736                       /usr/lib/locale/C.utf8/LC_MEASUREMENT
78ae07069000-78ae0706a000 r--p 00000000 08:30 2735                       /usr/lib/locale/C.utf8/LC_IDENTIFICATION
78ae0706a000-78ae0706c000 rw-p 00000000 00:00 0
78ae0706c000-78ae07070000 r--p 00000000 00:00 0                          [vvar]
78ae07070000-78ae07072000 r--p 00000000 00:00 0                          [vvar_vclock]
78ae07072000-78ae07074000 r-xp 00000000 00:00 0                          [vdso]
78ae07074000-78ae07075000 r--p 00000000 08:30 13931                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
78ae07075000-78ae070a4000 r-xp 00001000 08:30 13931                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
78ae070a4000-78ae070af000 r--p 00030000 08:30 13931                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
78ae070af000-78ae070b1000 r--p 0003b000 08:30 13931                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
78ae070b1000-78ae070b2000 rw-p 0003d000 08:30 13931                      /usr/lib/x86_64-linux-gnu/ld-linux-x86-64
.so.2
78ae070b2000-78ae070b3000 rw-p 00000000 00:00 0
7ffcfa58c000-7ffcfa5ad000 rw-p 00000000 00:00 0                          [stack]
ketana_sindhu@KetaZone:~$
```